Healthcare Data Cleaning and Understanding

Dataset exploration, cleaning and basic analysis

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Objective. Explore a healthcare dataset (age, gender, disease, medication, dosage), find and fix data quality problems, and report patient count, common diseases and average age. Tools: Python (pandas, matplotlib). Dataset: a fictional 106-row sample created for practice; no real patient data.

1. UNDERSTANDING THE DATASET

Column	Meaning	Type	Notes
Patient_ID	Unique patient code	Identifier (text)	Used to detect duplicates
Age	Age in years	Numerical (discrete)	Arrived as text in places, e.g. '45 years'
Gender	Male / Female	Categorical (nominal)	Mixed spellings: M, male, FEMALE
Disease	Diagnosis	Categorical (nominal)	Mixed capitals and extra spaces
Medication	Drug name	Categorical (nominal)	Generic drug names
Dosage	Dose per administration	Numerical (continuous)	Text such as '500 mg' and '500mg'; converted to Dosage_mg

Shape before cleaning: **106 rows x 6 columns**.

2. PROBLEMS FOUND

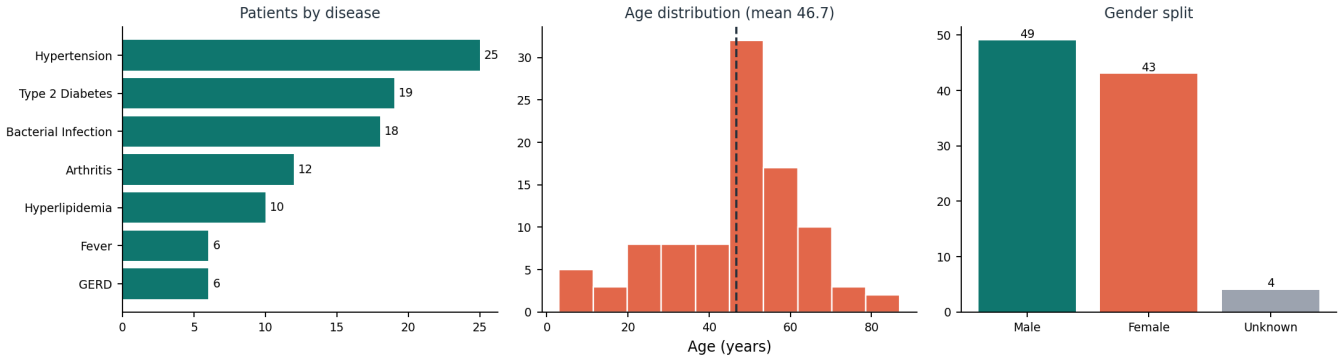
Problem	Count	Where
Exact duplicate rows	6	Whole records repeated
Missing Age	7	Age column
Missing Gender	4	Gender column
Missing Disease / Medication	2 / 2	Key clinical fields
Missing Dosage	5	Dosage column
Impossible ages (150 and -5)	2	Age column
Inconsistent text	many	Gender spellings, disease capitals, dosage units

3. CLEANING STEPS AND REASONS

No.	What was done	Why
1	Removed exact duplicate rows (6 removed)	A repeated patient would be counted twice
2	Stripped spaces; standardised Gender to Male/Female and fixed disease and drug capitalisation	So the same item is not counted as different values
3	Extracted numbers from Age and Dosage text; stored dose as Dosage_mg (integer)	Numbers are needed for averages
4	Set impossible ages (below 0 or above 110) to missing	They are data-entry errors
5	Dropped 4 rows missing Disease or Medication	Cannot analyse a case without its core clinical fields
6	Filled missing Age (8) with the median (49); median resists outliers	Keeps the record without distorting the average much
7	Filled missing Gender (4) as 'Unknown'	Guessing gender would be wrong
8	Filled missing Dosage (5) with the most common dose of that same medication	Dose depends on the drug
9	Checked for repeated Patient_IDs (0 found)	Confirms one row per patient

4. BASIC ANALYSIS (AFTER CLEANING)

Measure	Result
Rows before / after cleaning	106 / 96
Total patient count	96
Average age	46.7 years (median 49; range 3 to 87)
Most common disease	Hypertension (25 patients, 26%)
Top 3 diseases	Hypertension (25), Type 2 Diabetes (19), Bacterial Infection (18)
Most used medication	Metformin (19 patients)
Gender split	Male 49, Female 43, Unknown 4
Average age by gender	Female 46.2, Male 47.0



Disease count table.

Disease	Patients	Average age
Hypertension	25	56.7
Type 2 Diabetes	19	51.5
Bacterial Infection	18	26.8
Arthritis	12	57.6
Hyperlipidemia	10	53.7
Fever	6	26.5
GERD	6	35.7

5. FINDINGS

• The cleaned dataset has **96 unique patients**; cleaning removed 10 rows (6 duplicates and 4 with missing key fields).
• **Hypertension** is the most common disease, followed by Type 2 Diabetes and Bacterial Infection. Chronic conditions (hypertension, diabetes, arthritis, hyperlipidemia) make up most cases.
• The average age is **46.7 years**. Patients with chronic diseases are older (about 52 to 58) than those with infections or fever (about 27).
• Using the median for missing ages and the most common dose per drug kept the sample size without inventing unusual values.
• Limitation: a few values were filled in, so results are approximate. In real practice the source record should be checked first.

Fictional dataset for internship practice. Files: healthcare_dataset_raw.csv, healthcare_dataset_cleaned.csv, healthcare_cleaning.py.